Step-III : $w_3 = u_3 - \frac{\langle u_3, w_1\rangle}{\langle w_1, w_1\rangle} w_1 - \frac{\langle u_3, w_2\rangle}{\langle w_2, w_2\rangle} \cdot w_2$

$$\vdots$$

$$w_n = u_n - \sum_{i=1}^{n-1} \frac{\langle u_n, w_i\rangle}{\langle w_i, w_i\rangle} w_i$$

This process is known as Gram-Schmidt Orthogonalization

Ex:- $\{(1,0,1), (1,1,1), (-1,1,0)\}$

$w_1 = (1,0,1)$

$w_2 = u_2 - c_1 w_1 = u_2 - \frac{\langle u_2, w_1\rangle}{\langle w_1, w_1\rangle} w_1$

$= (1,1,1) - \frac{2}{2}(1,0,1) = (0,1,0)$

w_1 and w_2 are orthogonal.

$w_3 = u_3 - c_1 w_1 - c_2 w_2$

$= u_3 - \frac{\langle u_3, w_1\rangle}{\langle w_1, w_1\rangle} w_1 - \frac{\langle u_3, w_2\rangle}{\langle w_2, w_2\rangle} w_2$

$= (-1,1,0) - \frac{(-1)}{2}(1,0,1) - \frac{1}{1}(0,1,0)$

$= (-1,1,0) + \left(\frac{1}{2}, 0, \frac{1}{2}\right) + (0,-1,0)$

$= \left(\frac{-1}{2}, 0, \frac{1}{2}\right)$

$\therefore \left\{(1,0,1), (0,1,0), \left(\frac{-1}{2}, 0, \frac{1}{2}\right)\right\}$ is an orthogonal basis of V.

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